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OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CIVIL, EEE, ECE, BME, CSE) I-Semester (MAIN)
Examinations JAN/FEB-2025

APPLIED PHYSICS

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	List the properties of Matter Waves.	2	1	1
b)	State the Uncertainty Principle.	2	2	1
c)	Define Displacement Current.	2	1	2
d)	Determine the wavelength of radiation of given out by a laser with an energy of 3 eV? (h=6.63 x 10^{-34}J and c = 3 x 10^{8} m/s).	2	5	3
e)	Draw the Characteristic Curves of LED.	2	6	4
f)	Why Carbon Nanotubes are significant.	2	1	5
2. a)	Deduce an expression for quantized energy levels of a particle in 1-Dimension box.	6	6	1
b)	Verify the dual nature of Matter by Davision and Gernmer's experiment.	6	5	1
3. a)	State the Poynting Vector. Deduce an expression for Poynting Vector.	6	6	2
b)	Estimate the conditions for maxima and minima of a Diffraction Grating.	6	6	2
4. a)	Discuss the need of Stimulated Emission process.	6	6	3
b)	Elaborate Fibre Drawing Process by Double Crucible Method.	6	6	3
5. a)	Analyze Band formation for solid materials based on Kroing-Penney Model .	6	4	4
b)	What is superconductor. Distinguish between Type-1 and Type-2 Superconductors .	6	6	4

(P.T.O.)

6. a)	Elaborate Surface to Volume Ratio and Quntum Confinement Effect for nanomaterials.	6	6
b)	Discuss the construction and working of Scanning Electron Microscopy.	6	6
7. a)	Discuss Hall Effect and Deduce an expression for Hall Coefficient.	6	4
b)	Discuss the Sol-Gel method of preparation of nanomaterials.	6	6